

Fourier Analysis of Galactic Bars

1. Fourier Definition

For a galaxy image in polar coordinates (r, θ) :

$$A_m(r) = \frac{1}{I_0(r)} \int_0^{2\pi} I(r, \theta) e^{-im\theta} d\theta \quad (1)$$

Discrete Form

$$A_m(r) = \frac{\sum I_i(r) \cos(m\theta_i)}{\sum I_i(r)} \quad (2)$$

$$B_m(r) = \frac{\sum I_i(r) \sin(m\theta_i)}{\sum I_i(r)} \quad (3)$$

Amplitude

$$|A_m(r)| = \sqrt{A_m^2 + B_m^2} \quad (4)$$

Bar Mode ($m = 2$)

$$A_2(r) = \frac{\sum I_i \cos(2\theta_i)}{\sum I_i} \quad (5)$$

$$B_2(r) = \frac{\sum I_i \sin(2\theta_i)}{\sum I_i} \quad (6)$$

$$\frac{A_2}{A_0} = \sqrt{A_2^2 + B_2^2} \quad (7)$$

This quantity represents the **bar strength**.

2. Improvements for Real Data

Deprojection

Correct for inclination and rotate the galaxy to face-on orientation.

Elliptical Annuli

$$r^2 = x'^2 + \frac{y'^2}{q^2} \quad (8)$$

where $q = b/a$.

Phase

$$\phi_2(r) = \frac{1}{2} \tan^{-1} \left(\frac{B_2}{A_2} \right) \quad (9)$$

Bar region corresponds to approximately constant phase.

3. Bar Size Estimation

$$R_{\text{bar}} = R(A_2^{\text{max}}) \quad (10)$$

4. Error Estimation

$$\sigma_{R_{\text{bar}}} = \sqrt{\sigma_{\text{method}}^2 + \sigma_{\text{MC}}^2 + \sigma_{\text{bin}}^2} \quad (11)$$

5. Dynamical Ratio

$$\mathcal{R} = \frac{R_{\text{corot}}}{R_{\text{bar}}} \quad (12)$$

$$\sigma_{\mathcal{R}} = \mathcal{R} \sqrt{\left(\frac{\sigma_{R_{\text{corot}}}}{R_{\text{corot}}} \right)^2 + \left(\frac{\sigma_{R_{\text{bar}}}}{R_{\text{bar}}} \right)^2} \quad (13)$$

6. Interpretation

$\mathcal{R}$	Meaning
1.0 – 1.4	Fast bar
> 1.4	Slow bar

Conclusion

The $m = 2$ Fourier mode (A_2) captures the dominant bar structure in galaxies. Its amplitude provides a direct measure of bar strength, while its radial behavior allows estimation of the bar length and its dynamical properties.

References

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